Mira Kaplan

linkedin.com/in/mira-kaplan-cs

Email : mkaplan@uchicago.edu

Mobile : +1-773-555-0148

Education

University of Chicago Chicago, IL

B.S. Computer Science; GPA: 3.94 *Sep. 2024 – Jun. 2028*

- Coursework: Operating Systems, Computer Networks, Parallel Computing, Databases, Algorithms; Quantitative Portfolio Management and Algorithmic Trading (Financial Mathematics)
- Teaching Assistant, Introduction to Computer Systems, 3 quarters **Graduating Jun. 2028; available May – Aug. 2027, Chicago**

Experience

IMC Trading Chicago, IL

Software Engineer Intern, Options Trading Technology *Jun. 2026 – Aug. 2026*

- Built a validation framework letting traders define their own guardrails on live option pricing model output, from a small expression language through evaluation and alert routing; shipped to production in week three and adopted by two desks.
- Cut tick-to-quote latency 22% on the equity options path by replacing a shared-mutex conflation buffer with a per-instrument single-producer ring, measured against a replayed 90-minute captured session rather than synthetic load.

Cboe Global Markets Chicago, IL

Software Engineer Intern, Market Data *Jun. 2025 – Aug. 2025*

- Built the capture and replay harness for a multicast options feed, persisting 4TB/day of raw packets with nanosecond hardware timestamps and replaying any window bit for bit for regression testing.
- Turned a recurring gap-recovery incident into an automated pre-open check with the market operations team, closing a class of tickets they had been resolving by hand every week.

Projects

Column-Oriented Binary Format for Tick Data Capture

C++, Python C API *Feb. 2026*

- Designed a column-oriented capture format for raw exchange messages with a Python C extension for zero-copy reads, cutting storage 11x and full-day query time from 40 minutes to under 3 across a 2B-message corpus.

Distributed Backtest Scheduler

Rust, Python, Kubernetes *Apr. 2026*

- Built a scheduler fanning research backtests across a bare-metal pool with deterministic replay and content-addressed result caching, cutting a full parameter sweep from 6 hours to 25 minutes and making every rerun reproducible.

Lock-Free Limit Order Book and Matching Engine

C++20, Lock-Free Data Structures *Sep. 2025*

- Implemented price-time priority matching over a cache-aligned intrusive book fed by a single-producer ring buffer, sustaining 1M+ orders/sec at p99 under 800ns, validated against 10M+ replayed order events.

Leadership & Involvement

UChicago Trading Competition Chicago, IL

Case Author and Technology Lead *Sep. 2025 – Present*

- Wrote the market making case and built the simulated exchange it runs on, matching 300+ competitor algorithms over two days; coordinated engineers from four sponsoring proprietary trading firms as case reviewers.

Skills

Languages: C++ (17/20), Python, Java, Rust, C#, SQL

Systems: Lock-Free Concurrency, Cache-Aware Data Layout, UDP Multicast & TCP, Low-Latency Data Pipelines, Distributed Systems

Tools: Linux, GDB, perf, CMake, Kafka, PostgreSQL, Kubernetes, Docker, Git